Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/09/2015
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan Ph.D. (University of Hawaii at Manoa) Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and		

	minimize pollutants. This effort is a collaborative effort between the Principal Investigator, the three Co-Investigators, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments • The research team published 6 conference papers and abstracts during this period. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, chemical kinetics, and laser-induced breakdown spectroscopy (LIBS). • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 5 s drop facility. Additional Accomplishments W. Wu, P.B. Sunderland, B.-H. Chao, D.L. Urban, R.L. Axelbaum, Recent Advances in Flame Design, A Spaceflight Combustion Experiment, Abstract, ASGSR, Alexandria, VA, 2015. Flame Design is a spaceflight combustion experiment that will examine several unusual aspects of spherical diffusion flames fueled by gases. Its emphasis is on soot formation and flame extinction. This experiment, and associated ground-based work, attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints, design flames that best optimize efficiency and minimize pollutants. This presentation will introduce the overall concepts of Flame Design. It will summarize observations in the NASA Glenn 2.2 s and 5 s drop facilities, which have yielded important insights into soot formation and flame extinction behavior. Detailed numerical modeling of spherical diffusion flames have aided the understanding of the observations. Recent advances in Laser-Induced Breakdown Spectroscopy (LIBS) will be discussed. LIBS allows the C/O atom ratio to be measured in flames. This diagnostic is well suited to use in spaceflight experiments.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Wu W, Axelbaum RL. "Simultaneous Measurement of Conserved Scalars in Flames Using LIBS." SciX 2015, The Great Scientific Exchange, Providence, Rhode Island, September 27-October 2, 2015. SciX 2015, The Great Scientific Exchange, Providence, Rhode Island, September 27-October 2, 2015. , Oct-2015
Abstracts for Journals and Proceedings	Wu W, Sunderland PB, Chao B-H, Urban DL, Axelbaum RL. "Recent Advances in Flame Design, A Spaceflight Combustion Experiment." To be presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Wu W, Yablonsky G, Axelbaum RL. "New Observation of Water Gas Shift Equilibrium in Flames." 9th International Conference on Chemical Kinetics, Ghent, Belgium, June 28-July 2, 2015. 9th International Conference on Chemical Kinetics, Ghent, Belgium, June 28-July 2, 2015. , Jul-2015
Abstracts for Journals and Proceedings	Lecoustre VR, Sunderland PB, Chao B-H, Axelbaum RL. "Study of Temperature Effects on Soot Precursor Formation in Nonbuoyant, Laminar Ethylene Premixed Flames." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao B-H, Axelbaum RL. "Numerical Investigations of Temperature Effects on Soot Precursors in Ethylene Premixed Flames." 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. Paper IE09. , May-2015
Papers from Meeting Proceedings	Wu W, Axelbaum RL. "Interpreting Diffusion Flame Structure by Measuring C/O Ratio Using Laser-Induced Breakdown Spectroscopy." 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20, 2015. Paper IA20. , May-2015